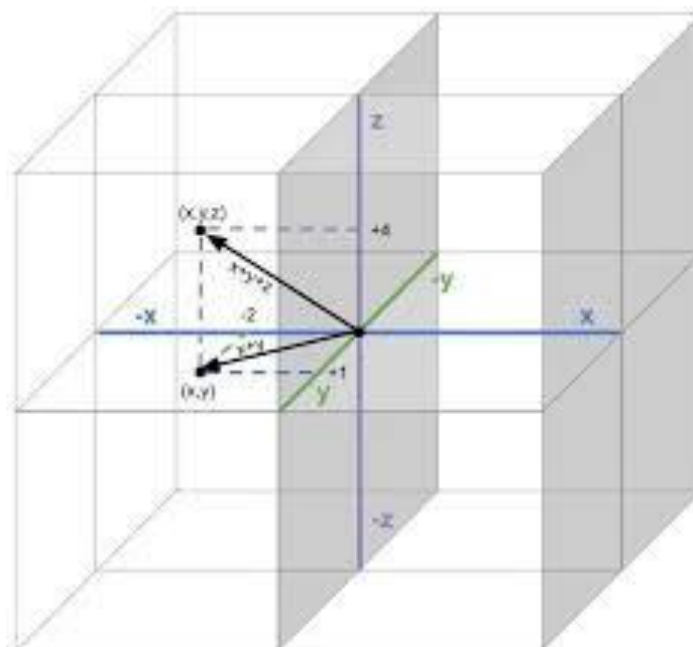


Linear Algebra

Math 1057

Instructor

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Lecture 14

Vector Spaces

14.1 Vector Spaces

When you read/hear the word **vector** you may immediately think of two points in $\mathbb{R}^2$ (or $\mathbb{R}^3$) connected by an arrow. Mathematically speaking, a **vector** is just an element of a **vector space**. This then begs the question: What is a **vector space**? Roughly speaking, a vector space is **a set of objects that can be added and multiplied by scalars**. You have already worked with several types of vector spaces. Examples of vector spaces that you have already encountered are:

1. the set $\mathbb{R}^n$,
2. the set of all $n \times n$ matrices,
3. the set of all functions from $[a, b]$ to $\mathbb{R}$, and
4. the set of all sequences.

In all of these sets, there is an operation of “addition” and “multiplication by scalars”. Let’s formalize then exactly what we mean by a vector space.

Definition 14.1: A **vector space** is a set V of objects, called **vectors**, on which two operations called **addition** and **scalar multiplication** have been defined satisfying the following properties. If $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are in V and if $\alpha, \beta \in \mathbb{R}$ are scalars:

- (1) The sum $\mathbf{u} + \mathbf{v}$ is in V . (closure under addition)
- (2) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (addition is commutative)
- (3) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (addition is associativity)
- (4) There is a vector in V called the **zero** vector, denoted by $\mathbf{0}$, satisfying $\mathbf{v} + \mathbf{0} = \mathbf{v}$.
- (5) For each $\mathbf{v}$ there is a vector $-\mathbf{v}$ in V such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.

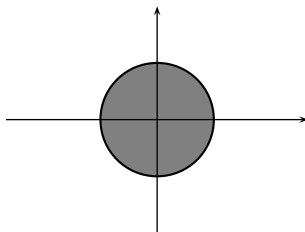
- (6) The scalar multiple of $\mathbf{v}$ by α , denoted $\alpha\mathbf{v}$, is in $\mathbf{V}$. (closure under scalar multiplication)
- (7) $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$
- (8) $(\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$
- (9) $\alpha(\beta\mathbf{v}) = (\alpha\beta)\mathbf{v}$
- (10) $1\mathbf{v} = \mathbf{v}$

It can be shown that $0 \cdot \mathbf{v} = \mathbf{0}$ for any vector $\mathbf{v}$ in $\mathbf{V}$. To better understand the definition of a vector space, we first consider a few elementary examples.

Example 14.2. Let $\mathbf{V}$ be the unit disc in $\mathbb{R}^2$:

$$\mathbf{V} = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$$

Is $\mathbf{V}$ a vector space?

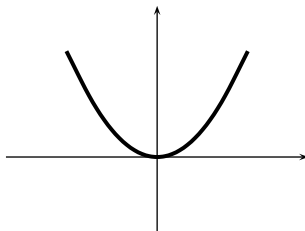


Solution. The circle is not closed under scalar multiplication. For example, take $\mathbf{u} = (1, 0) \in \mathbf{V}$ and multiply by say $\alpha = 2$. Then $\alpha\mathbf{u} = (2, 0)$ is not in $\mathbf{V}$. Therefore, property (6) of the definition of a vector space fails, and consequently the unit disc is not a vector space. $\square$

Example 14.3. Let $\mathbf{V}$ be the graph of the quadratic function $f(x) = x^2$:

$$\mathbf{V} = \{(x, y) \in \mathbb{R}^2 \mid y = x^2\}.$$

Is $\mathbf{V}$ a vector space?

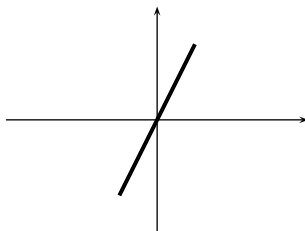


Solution. The set $\mathbf{V}$ is not closed under scalar multiplication. For example, $\mathbf{u} = (1, 1)$ is a point in $\mathbf{V}$ but $2\mathbf{u} = (2, 2)$ is not. You may also notice that $\mathbf{V}$ is not closed under addition either. For example, both $\mathbf{u} = (1, 1)$ and $\mathbf{v} = (2, 4)$ are in $\mathbf{V}$ but $\mathbf{u} + \mathbf{v} = (3, 5)$ and $(3, 5)$ is not a point on the parabola $\mathbf{V}$. Therefore, the graph of $f(x) = x^2$ is not a vector space. $\square$

Example 14.4. Let V be the graph of the function $f(x) = 2x$:

$$V = \{(x, y) \in \mathbb{R}^2 \mid y = 2x\}.$$

Is V a vector space?



Solution. We will show that V is a vector space. First, we verify that V is closed under addition. We first note that an arbitrary point in V can be written as $\mathbf{u} = (x, 2x)$. Let then $\mathbf{u} = (a, 2a)$ and $\mathbf{v} = (b, 2b)$ be points in V . Then

$$\mathbf{u} + \mathbf{v} = (a + b, 2a + 2b) = (a + b, 2(a + b)).$$

Therefore V is closed under addition. Verify that V is closed under scalar multiplication:

$$\alpha \mathbf{u} = \alpha(a, 2a) = (\alpha a, \alpha 2a) = (\alpha a, 2(\alpha a)).$$

Therefore V is closed under scalar multiplication. There is a zero vector $\mathbf{0} = (0, 0)$ in V :

$$\mathbf{u} + \mathbf{0} = (a, 2a) + (0, 0) = (a, 2a).$$

All the other properties of a vector space can be verified to hold; for example, addition is commutative and associative in V because addition in $\mathbb{R}^2$ is commutative/associative, etc. Therefore, the graph of the function $f(x) = 2x$ is a vector space. $\square$

The following example is important (it will appear frequently) and is our first example of what we could say is an “abstract vector space”. To emphasize, a vector space is a set that comes equipped with an operation of addition and scalar multiplication and these two operations satisfy the list of properties above.

Example 14.5. Let $V = \mathbb{P}_n[t]$ be the set of all polynomials in the variable t and of degree at most n :

$$\mathbb{P}_n[t] = \left\{ a_0 + a_1 t + a_2 t^2 + \cdots + a_n t^n \mid a_0, a_1, \dots, a_n \in \mathbb{R} \right\}.$$

Is V a vector space?

Solution. Let $\mathbf{u}(t) = u_0 + u_1 t + \cdots + u_n t^n$ and let $\mathbf{v}(t) = v_0 + v_1 t + \cdots + v_n t^n$ be polynomials in V . We define the addition of $\mathbf{u}$ and $\mathbf{v}$ as the new polynomial $(\mathbf{u} + \mathbf{v})$ as follows:

$$(\mathbf{u} + \mathbf{v})(t) = \mathbf{u}(t) + \mathbf{v}(t) = (u_0 + v_0) + (u_1 + v_1)t + \cdots + (u_n + v_n)t^n.$$

Then $\mathbf{u} + \mathbf{v}$ is a polynomial of degree at most n and thus $(\mathbf{u} + \mathbf{v}) \in \mathbb{P}_n[t]$, and therefore this shows that $\mathbb{P}_n[t]$ is closed under addition. Now let α be a scalar, define a new polynomial $(\alpha\mathbf{u})$ as follows:

$$(\alpha\mathbf{u})(t) = (\alpha u_0) + (\alpha u_1)t + \cdots + (\alpha u_n)t^n$$

Then $(\alpha\mathbf{u})$ is a polynomial of degree at most n and thus $(\alpha\mathbf{u}) \in \mathbb{P}_n[t]$; hence, $\mathbb{P}_n[t]$ is closed under scalar multiplication. The $\mathbf{0}$ vector in $\mathbb{P}_n[t]$ is the zero polynomial $\mathbf{0}(t) = 0$. One can verify that all other properties of the definition of a vector space also hold; for example, addition is commutative and associative, etc. Thus $\mathbb{P}_n[t]$ is a vector space. $\square$

Example 14.6. Let $\mathbf{V} = M_{m \times n}$ be the set of all $m \times n$ matrices. Under the usual operations of addition of matrices and scalar multiplication, is $M_{n \times m}$ a vector space?

Solution. Given matrices $\mathbf{A}, \mathbf{B} \in M_{m \times n}$ and a scalar α , we defined the sum $\mathbf{A} + \mathbf{B}$ by adding entry-by-entry, and $\alpha\mathbf{A}$ by multiplying each entry of $\mathbf{A}$ by α . It is clear that the space $M_{m \times n}$ is closed under these two operations. The $\mathbf{0}$ vector in $M_{m \times n}$ is the matrix of size $m \times n$ having all entries equal to zero. It can be verified that all other properties of the definition of a vector space also hold. Thus, the set $M_{m \times n}$ is a vector space. $\square$

Example 14.7. The n -dimensional Euclidean space $\mathbf{V} = \mathbb{R}^n$ under the usual operations of addition and scalar multiplication is vector space.

Example 14.8. Let $\mathbf{V} = C[a, b]$ denote the set of functions with domain $[a, b]$ and co-domain $\mathbb{R}$ that are continuous. Is $\mathbf{V}$ a vector space?

14.2 Subspaces of Vector Spaces

Frequently, one encounters a vector space $\mathbf{W}$ that is a subset of a larger vector space $\mathbf{V}$. In this case, we would say that $\mathbf{W}$ is a subspace of $\mathbf{V}$. Below is the formal definition.

Definition 14.9: Let $\mathbf{V}$ be a vector space. A subset $\mathbf{W}$ of $\mathbf{V}$ is called a **subspace** of $\mathbf{V}$ if it satisfies the following properties:

- (1) The zero vector of $\mathbf{V}$ is also in $\mathbf{W}$.
- (2) $\mathbf{W}$ is closed under addition, that is, if $\mathbf{u}$ and $\mathbf{v}$ are in $\mathbf{W}$ then $\mathbf{u} + \mathbf{v}$ is in $\mathbf{W}$.
- (3) $\mathbf{W}$ is closed under scalar multiplication, that is, if $\mathbf{u}$ is in $\mathbf{W}$ and α is a scalar then $\alpha\mathbf{u}$ is in $\mathbf{W}$.

Example 14.10. Let $\mathbf{W}$ be the graph of the function $f(x) = 2x$:

$$\mathbf{W} = \{(x, y) \in \mathbb{R}^2 \mid y = 2x\}.$$

Is $\mathbf{W}$ a subspace of $\mathbf{V} = \mathbb{R}^2$?

Solution. If $x = 0$ then $y = 2 \cdot 0 = 0$ and therefore $\mathbf{0} = (0, 0)$ is in $\mathbf{W}$. Let $\mathbf{u} = (a, 2a)$ and $\mathbf{v} = (b, 2b)$ be elements of $\mathbf{W}$. Then

$$\mathbf{u} + \mathbf{v} = (a, 2a) + (b, 2b) = (a + b, 2a + 2b) = (\underbrace{a + b}_x, 2\underbrace{(a + b)}_x).$$

Because the x and y components of $\mathbf{u} + \mathbf{v}$ satisfy $y = 2x$ then $\mathbf{u} + \mathbf{v}$ is inside in $\mathbf{W}$. Thus, $\mathbf{W}$ is closed under addition. Let α be any scalar and let $\mathbf{u} = (a, 2a)$ be an element of $\mathbf{W}$. Then

$$\alpha\mathbf{u} = (\alpha a, \alpha 2a) = (\underbrace{\alpha a}_x, 2\underbrace{(\alpha a)}_x)$$

Because the x and y components of $\alpha\mathbf{u}$ satisfy $y = 2x$ then $\alpha\mathbf{u}$ is an element of $\mathbf{W}$, and thus $\mathbf{W}$ is closed under scalar multiplication. All three conditions of a subspace are satisfied for $\mathbf{W}$ and therefore $\mathbf{W}$ is a subspace of $\mathbf{V}$. $\square$

Example 14.11. Let $\mathbf{W}$ be the first quadrant in $\mathbb{R}^2$:

$$\mathbf{W} = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0\}.$$

Is $\mathbf{W}$ a subspace?

Solution. The set $\mathbf{W}$ contains the zero vector and the sum of two vectors in $\mathbf{W}$ is again in $\mathbf{W}$; you may want to verify this explicitly as follows: if $\mathbf{u}_1 = (x_1, y_1)$ is in $\mathbf{W}$ then $x_1 \geq 0$ and $y_1 \geq 0$, and similarly if $\mathbf{u}_2 = (x_2, y_2)$ is in $\mathbf{W}$ then $x_2 \geq 0$ and $y_2 \geq 0$. Then the sum $\mathbf{u}_1 + \mathbf{u}_2 = (x_1 + x_2, y_1 + y_2)$ has components $x_1 + x_2 \geq 0$ and $y_1 + y_2 \geq 0$ and therefore $\mathbf{u}_1 + \mathbf{u}_2$ is in $\mathbf{W}$. However, $\mathbf{W}$ is not closed under scalar multiplication. For example if $\mathbf{u} = (1, 1)$ and $\alpha = -1$ then $\alpha\mathbf{u} = (-1, -1)$ is not in $\mathbf{W}$ because the components of $\alpha\mathbf{u}$ are clearly not non-negative. $\square$

Example 14.12. Let $\mathbf{V} = M_{n \times n}$ be the vector space of all $n \times n$ matrices. We define the **trace** of a matrix $\mathbf{A} \in M_{n \times n}$ as the sum of its diagonal entries:

$$\text{tr}(\mathbf{A}) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Let $\mathbf{W}$ be the set of all $n \times n$ matrices whose trace is zero:

$$\mathbf{W} = \{\mathbf{A} \in M_{n \times n} \mid \text{tr}(\mathbf{A}) = 0\}.$$

Is $\mathbf{W}$ a subspace of $\mathbf{V}$?

Solution. If $\mathbf{0}$ is the $n \times n$ zero matrix then clearly $\text{tr}(\mathbf{0}) = 0$, and thus $\mathbf{0} \in M_{n \times n}$. Suppose that $\mathbf{A}$ and $\mathbf{B}$ are in $\mathbf{W}$. Then necessarily $\text{tr}(\mathbf{A}) = 0$ and $\text{tr}(\mathbf{B}) = 0$. Consider the matrix $\mathbf{C} = \mathbf{A} + \mathbf{B}$. Then

$$\begin{aligned} \text{tr}(\mathbf{C}) &= \text{tr}(\mathbf{A} + \mathbf{B}) = (a_{11} + b_{11}) + (a_{22} + b_{22}) + \cdots + (a_{nn} + b_{nn}) \\ &= (a_{11} + \cdots + a_{nn}) + (b_{11} + \cdots + b_{nn}) \\ &= \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B}) \\ &= 0 \end{aligned}$$

Therefore, $\text{tr}(\mathbf{C}) = 0$ and consequently $\mathbf{C} = \mathbf{A} + \mathbf{B} \in \mathbf{W}$, in other words, $\mathbf{W}$ is closed under addition. Now let α be a scalar and let $\mathbf{C} = \alpha\mathbf{A}$. Then

$$\text{tr}(\mathbf{C}) = \text{tr}(\alpha\mathbf{A}) = (\alpha a_{11}) + (\alpha a_{22}) + \cdots + (\alpha a_{nn}) = \alpha \text{tr}(\mathbf{A}) = 0.$$

Thus, $\text{tr}(\mathbf{C}) = 0$, that is, $\mathbf{C} = \alpha\mathbf{A} \in \mathbf{W}$, and consequently $\mathbf{W}$ is closed under scalar multiplication. Therefore, the set $\mathbf{W}$ is a subspace of $\mathbf{V}$. $\square$

Example 14.13. Let $\mathbf{V} = \mathbb{P}_n[t]$ and consider the subset $\mathbf{W}$ of $\mathbf{V}$:

$$\mathbf{W} = \{u \in \mathbb{P}_n[t] \mid u'(1) = 0\}$$

In other words, $\mathbf{W}$ consists of polynomials of degree n in the variable t whose derivative at $t = 1$ is zero. Is $\mathbf{W}$ a subspace of $\mathbf{V}$?

Solution. The zero polynomial $\mathbf{0}(t) = 0$ clearly has derivative at $t = 1$ equal to zero, that is, $\mathbf{0}'(1) = 0$, and thus the zero polynomial is in $\mathbf{W}$. Now suppose that $\mathbf{u}(t)$ and $\mathbf{v}(t)$ are two polynomials in $\mathbf{W}$. Then, $\mathbf{u}'(1) = 0$ and also $\mathbf{v}'(1) = 0$. To verify whether or not $\mathbf{W}$ is closed under addition, we must determine whether the sum polynomial $(\mathbf{u} + \mathbf{v})(t)$ has a derivative at $t = 1$ equal to zero. From the rules of differentiation, we compute

$$(\mathbf{u} + \mathbf{v})'(1) = \mathbf{u}'(1) + \mathbf{v}'(1) = 0 + 0.$$

Therefore, the polynomial $(\mathbf{u} + \mathbf{v})$ is in $\mathbf{W}$, and thus $\mathbf{W}$ is closed under addition. Now let α be any scalar and let $\mathbf{u}(t)$ be a polynomial in $\mathbf{W}$. Then $\mathbf{u}'(1) = 0$. To determine whether or not the scalar multiple $\alpha\mathbf{u}(t)$ is in $\mathbf{W}$ we must determine if $\alpha\mathbf{u}(t)$ has a derivative of zero at $t = 1$. Using the rules of differentiation, we compute that

$$(\alpha\mathbf{u})'(1) = \alpha\mathbf{u}'(1) = \alpha \cdot 0 = 0.$$

Therefore, the polynomial $(\alpha\mathbf{u})(t)$ is in $\mathbf{W}$ and thus $\mathbf{W}$ is closed under scalar multiplication. All three properties of a subspace hold for $\mathbf{W}$ and therefore $\mathbf{W}$ is a subspace of $\mathbb{P}_n[t]$. $\square$

Example 14.14. Let $\mathbf{V} = \mathbb{P}_n[t]$ and consider the subset $\mathbf{W}$ of $\mathbf{V}$:

$$\mathbf{W} = \{u \in \mathbb{P}_n[t] \mid u(2) = -1\}$$

In other words, $\mathbf{W}$ consists of polynomials of degree n in the variable t whose value $t = 2$ is -1 . Is $\mathbf{W}$ a subspace of $\mathbf{V}$?

Solution. The zero polynomial $\mathbf{0}(t) = 0$ clearly does not equal -1 at $t = 2$. Therefore, $\mathbf{W}$ does not contain the zero polynomial and, because all three conditions of a subspace must be satisfied for $\mathbf{W}$ to be a subspace, then $\mathbf{W}$ is not a subspace of $\mathbb{P}_n[t]$. As an exercise, you may want to investigate whether or not $\mathbf{W}$ is closed under addition and scalar multiplication. $\square$

Example 14.15. A square matrix $\mathbf{A}$ is said to be **symmetric** if $\mathbf{A}^T = \mathbf{A}$. For example, here is a 3×3 symmetric matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & 5 \\ -3 & 5 & 7 \end{bmatrix}$$

Verify for yourself that we do indeed have that $\mathbf{A}^T = \mathbf{A}$. Let $\mathbf{W}$ be the set of all symmetric $n \times n$ matrices. Is $\mathbf{W}$ a subspace of $\mathbf{V} = M_{n \times n}$?

Example 14.16. For any vector space $\mathbf{V}$, there are two trivial subspaces in $\mathbf{V}$, namely, $\mathbf{V}$ itself is a subspace of $\mathbf{V}$ and the set consisting of the zero vector $\mathbf{W} = \{\mathbf{0}\}$ is a subspace of $\mathbf{V}$.

There is one particular way to generate a subspace of any given vector space $\mathbf{V}$ using the span of a set of vectors. Recall that we defined the span of a set of vectors in $\mathbb{R}^n$ but we can define the same notion on a general vector space $\mathbf{V}$.

Definition 14.17: Let $\mathbf{V}$ be a vector space and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ be vectors in $\mathbf{V}$. The **span** of $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is the set of all linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_p$:

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\} = \left\{ t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \dots + t_p\mathbf{v}_p \mid t_1, t_2, \dots, t_p \in \mathbb{R} \right\}.$$

We now show that the span of a set of vectors in $\mathbf{V}$ is a subspace of $\mathbf{V}$.

Theorem 14.18: If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ are vectors in $\mathbf{V}$ then $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is a subspace of $\mathbf{V}$.

Solution. Let $\mathbf{u} = t_1\mathbf{v}_1 + \dots + t_p\mathbf{v}_p$ and $\mathbf{w} = s_1\mathbf{v}_1 + \dots + s_p\mathbf{v}_p$ be two vectors in $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$. Then

$$\mathbf{u} + \mathbf{w} = (t_1\mathbf{v}_1 + \dots + t_p\mathbf{v}_p) + (s_1\mathbf{v}_1 + \dots + s_p\mathbf{v}_p) = (t_1 + s_1)\mathbf{v}_1 + \dots + (t_p + s_p)\mathbf{v}_p.$$

Therefore $\mathbf{u} + \mathbf{w}$ is also in the span of $\mathbf{v}_1, \dots, \mathbf{v}_p$. Now consider $\alpha\mathbf{u}$:

$$\alpha\mathbf{u} = \alpha(t_1\mathbf{v}_1 + \dots + t_p\mathbf{v}_p) = (\alpha t_1)\mathbf{v}_1 + \dots + (\alpha t_p)\mathbf{v}_p.$$

Therefore, $\alpha\mathbf{u}$ is in the span of $\mathbf{v}_1, \dots, \mathbf{v}_p$. Lastly, since $0\mathbf{v}_1 + 0\mathbf{v}_2 + \dots + 0\mathbf{v}_p = \mathbf{0}$ then the zero vector $\mathbf{0}$ is in the span of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$. Therefore, $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is a subspace of $\mathbf{V}$. $\square$

Given a general subspace $\mathbf{W}$ of $\mathbf{V}$, if $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p$ are vectors in $\mathbf{W}$ such that

$$\text{span}\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p\} = \mathbf{W}$$

then we say that $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p\}$ is a **spanning set** of $\mathbf{W}$. Hence, every vector in $\mathbf{W}$ can be written as a linear combination of the vectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p$.

After this lecture you should know the following:

- what a vector space/subspace is
- be able to give some examples of vector spaces/subspaces
- that the span of a set of vectors in V is a subspace of V